

Instructions. (100 points) Sections 15.3–16.6. You have approximately 70 minutes. You may use a calculator. Show all work for credit. Several problems ask you to **set up but not evaluate**; for those, *full credit will be given only if your setup uses a coordinate system that makes the integral easy to evaluate*. Setups that are correct but unnecessarily difficult will not earn full credit.

(12^{pts}) 1. Evaluate $\iint_D xy \, dA$, where D is the portion of the disk $x^2 + y^2 \leq 4$ lying in the first quadrant.

(12^{pts}) 2. **Set up, but do not evaluate**, an integral for the surface area of the part of the upper hemisphere $z = \sqrt{4 - x^2 - y^2}$ that lies inside the cylinder $x^2 + y^2 = 1$. *Full credit requires a setup whose integrals can be evaluated easily.*

(12^{pts}) **3. Set up, but do not evaluate,** an integral for the volume of the solid bounded by the paraboloids $y = x^2 + z^2$ and $y = 4 - x^2 - z^2$. *Full credit requires a setup whose integral can be evaluated easily.*

(12^{pts}) **4. Set up, but do not evaluate,** an integral for the volume of the solid bounded above by the sphere $x^2 + y^2 + z^2 = 4$ and below by the cone $z = \sqrt{3(x^2 + y^2)}$. *Full credit requires a setup whose integral can be evaluated easily.*

(12^{pts}) **5.** Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y) = \langle y, x + y \rangle$ and C is the parabolic arc $\mathbf{r}(t) = \langle t, t^2 \rangle$, $0 \leq t \leq 1$.

(14^{pts}) **6.** Let $\mathbf{F}(x, y) = \langle 2xy + y, x^2 + x \rangle$, and let C be the closed path from $(0, 0)$ to $(1, 1)$ along $y = x^2$, then back from $(1, 1)$ to $(0, 0)$ along $y = x$.

(a) (4 pts) Is $\mathbf{F}$ conservative? If so, what does that imply about $\oint_C \mathbf{F} \cdot d\mathbf{r}$ before doing any computation?

(b) (6 pts) Verify your answer in (a) by applying Green's Theorem.

(c) (4 pts) Find a potential function $f(x, y)$ with $\nabla f = \mathbf{F}$.

(14pts) **7.** Use Green's Theorem with $\mathbf{F}(x, y) = \langle -\frac{1}{2}y, \frac{1}{2}x \rangle$ to prove that the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab .

(a) (6 pts) Compute $Q_x - P_y$ for this $\mathbf{F}$. What does Green's Theorem then tell you about the value of $\oint_C \mathbf{F} \cdot d\mathbf{r}$ when C is any simple closed curve in the plane oriented counterclockwise?

(b) (8 pts) Parameterize the ellipse and compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ directly. Use part (a) to conclude that the area enclosed by the ellipse is πab .

(12^{pts}) **8.** For each statement, decide whether it is **true** or **false**, and give one sentence of justification.

(a) (3 pts) If $\mathbf{F}$ is continuously differentiable on $\mathbb{R}^2$ and $P_y = Q_x$ at every point, then $\mathbf{F}$ is conservative.

(b) (3 pts) If $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for one particular simple closed curve C , then $\mathbf{F}$ is conservative.

(c) (3 pts) If $\operatorname{div} \mathbf{F} = 0$ at every point of a region, then $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve C in that region.

(d) (3 pts) Green's Theorem applies whenever C is a simple closed curve in the plane and P, Q are continuously differentiable on the region D enclosed by C .